

Task 16: Incident Response & Security Breach Simulation

Primary Tools

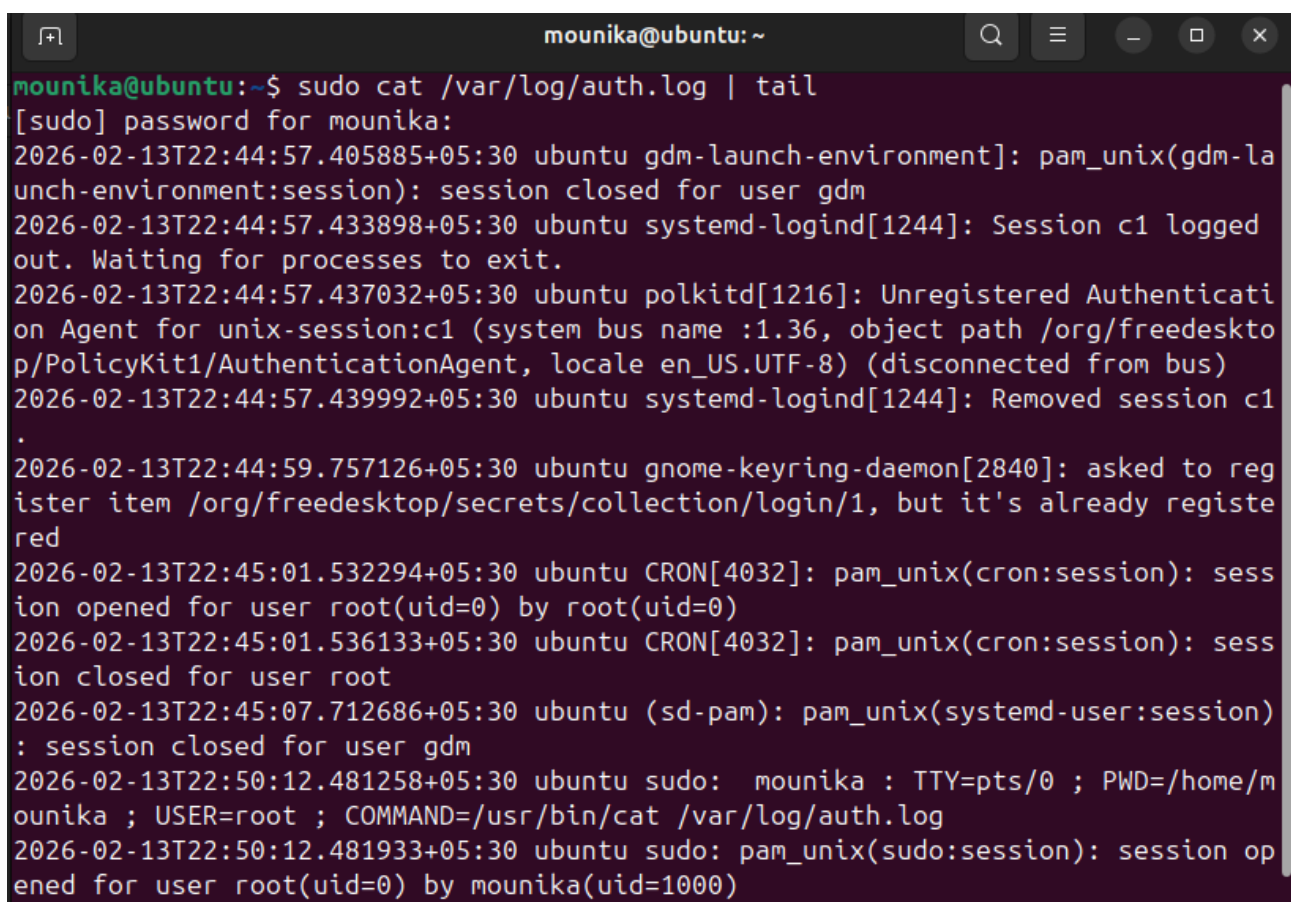
- Linux Authentication Logs
- OpenSSH Service (SSH Login Testing)

1.Incident Simulation

In this task, I simulated a basic security incident by generating multiple failed login attempts using incorrect credentials. This helped me understand how brute force login attacks appear in system logs and how security teams detect them.

First, I monitored authentication logs in Linux to observe login activities. Then, I created a test user account to safely simulate failed login attempts without affecting the system.

Screenshot : Monitoring authentication logs

A terminal window titled 'mounika@ubuntu: ~' with standard Ubuntu window controls. The user has executed the command 'sudo cat /var/log/auth.log | tail'. The output shows a series of log entries from the system's authentication logs, including session closures for 'gdm' and 'root', login attempts, and session openings for 'root' via 'cron' and 'sudo'.

```
mounika@ubuntu:~$ sudo cat /var/log/auth.log | tail
[sudo] password for mounika:
2026-02-13T22:44:57.405885+05:30 ubuntu gdm-launch-environment]: pam_unix(gdm-launch-environment:session): session closed for user gdm
2026-02-13T22:44:57.433898+05:30 ubuntu systemd-logind[1244]: Session c1 logged out. Waiting for processes to exit.
2026-02-13T22:44:57.437032+05:30 ubuntu polkitd[1216]: Unregistered Authentication Agent for unix-session:c1 (system bus name :1.36, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8) (disconnected from bus)
2026-02-13T22:44:57.439992+05:30 ubuntu systemd-logind[1244]: Removed session c1.
2026-02-13T22:44:59.757126+05:30 ubuntu gnome-keyring-daemon[2840]: asked to register item /org/freedesktop/secrets/collection/login/1, but it's already registered
2026-02-13T22:45:01.532294+05:30 ubuntu CRON[4032]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-02-13T22:45:01.536133+05:30 ubuntu CRON[4032]: pam_unix(cron:session): session closed for user root
2026-02-13T22:45:07.712686+05:30 ubuntu (sd-pam): pam_unix(systemd-user:session): session closed for user gdm
2026-02-13T22:50:12.481258+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/cat /var/log/auth.log
2026-02-13T22:50:12.481933+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
```

Screenshot : Creating the test user

```
mounika@ubuntu:~$ sudo adduser testuser
info: Adding user `testuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `testuser' (1001) ...
info: Adding new user `testuser' (1001) with group `testuser (1001)' ...
info: Creating home directory `/home/testuser' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for testuser
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `testuser' to supplemental / extra groups `users' ...
info: Adding user `testuser' to group `users' ...
mounika@ubuntu:~$
```

Screenshot : Failed login attempts in terminal

```
mounika@ubuntu:~$ sudo nano /etc/ssh/sshd_config
mounika@ubuntu:~$ sudo systemctl restart ssh
mounika@ubuntu:~$ ssh testuser@localhost
testuser@localhost's password:
Permission denied, please try again.
testuser@localhost's password:
Permission denied, please try again.
testuser@localhost's password:
testuser@localhost: Permission denied (publickey,password).
mounika@ubuntu:~$
```

2. Suspicious Activity Identification

To identify suspicious activity, I analyzed the system authentication logs located in `/var/log/auth.log`. Using log filtering commands, I identified multiple failed password attempts, which is a strong indicator of a brute force attack attempt.

I also used live log monitoring to observe login failures in real time.

Screenshot : Failed password logs

```
mounika@ubuntu:~$ sudo grep "Failed password" /var/log/auth.log
2026-02-13T22:55:36.289901+05:30 ubuntu sshd[6321]: Failed password for testuser
from 127.0.0.1 port 41862 ssh2
2026-02-13T22:55:55.242366+05:30 ubuntu sshd[6321]: message repeated 2 times: [
Failed password for testuser from 127.0.0.1 port 41862 ssh2]
2026-02-13T22:57:42.435135+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/m
ounika ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
mounika@ubuntu:~$
```

Screenshot : Live failed login logs updating

```
mounika@ubuntu:~$ sudo tail -f /var/log/auth.log
2026-02-13T22:55:34.394530+05:30 ubuntu sshd[6321]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=testuser
2026-02-13T22:55:36.289901+05:30 ubuntu sshd[6321]: Failed password for testuser from 127.0.0.1 port 41862 ssh2
2026-02-13T22:55:55.242366+05:30 ubuntu sshd[6321]: message repeated 2 times: [ Failed password for testuser from 127.0.0.1 port 41862 ssh2]
2026-02-13T22:55:56.553149+05:30 ubuntu sshd[6321]: Connection closed by authenticating user testuser 127.0.0.1 port 41862 [preauth]
2026-02-13T22:55:56.553446+05:30 ubuntu sshd[6321]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=testuser
2026-02-13T22:57:42.435135+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-02-13T22:57:42.436547+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-13T22:57:42.442007+05:30 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-13T22:58:27.230555+05:30 ubuntu sudo: mounika : TTY=pts/0 ; PWD=/home/mounika ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-02-13T22:58:27.231309+05:30 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mounika(uid=1000)
2026-02-13T22:58:42.582431+05:30 ubuntu sshd[6429]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=testuser
2026-02-13T22:58:44.623281+05:30 ubuntu sshd[6429]: Failed password for testuser from 127.0.0.1 port 49280 ssh2
2026-02-13T22:58:59.685836+05:30 ubuntu sshd[6429]: message repeated 2 times: [ Failed password for testuser from 127.0.0.1 port 49280 ssh2]
2026-02-13T22:58:59.685836+05:30 ubuntu sshd[6429]: Connection closed by authenticating user testuser 127.0.0.1 port 49280 [preauth]
2026-02-13T22:58:59.686072+05:30 ubuntu sshd[6429]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1 user=testuser
```

3. Incident Classification

From my analysis, I classified the incident as:

- **Attack Type:** Brute Force Login Attempt
- **Severity:** Medium

Since the attack was simulated locally and no system compromise occurred, it was treated as medium severity.

4. Incident Containment

To contain the incident, I locked the compromised/suspicious user account using Linux account security commands. This prevents further login attempts using that account.

Screenshot : Account lock confirmation

```
mounika@ubuntu:~$ sudo passwd -l testuser
passwd: password changed.
mounika@ubuntu:~$
```

Screenshot → Login blocked(even with correct password)

```
mounika@ubuntu:~$ ssh testuser@localhost
testuser@localhost's password:
Permission denied, please try again.
testuser@localhost's password:
Permission denied, please try again.
testuser@localhost's password:
testuser@localhost: Permission denied (publickey,password).
mounika@ubuntu:~$
```

5. Threat Removal and Root Cause Fix

To prevent similar attacks in the future, I installed Fail2Ban, which automatically blocks IP addresses that generate repeated failed login attempts. This is a common real-world defensive security tool.

6. System Restoration

After containment and prevention measures were applied, I restarted the SSH service and verified system stability. This ensured the system returned to a secure and operational state.

Screenshot : Fail2Ban active status

```
mounika@ubuntu:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset:
   Active: active (running) since Fri 2026-02-13 23:03:54 IST; 13s ago
     Docs: man:fail2ban(1)
    Main PID: 7531 (fail2ban-server)
       Tasks: 5 (limit: 9001)
      Memory: 67.8M (peak: 72.9M)
         CPU: 298ms
        CGroup: /system.slice/fail2ban.service
               └─7531 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Feb 13 23:03:54 ubuntu systemd[1]: Started fail2ban.service - Fail2Ban Service.
Feb 13 23:03:54 ubuntu fail2ban-server[7531]: 2026-02-13 23:03:54,867 fail2ban.>
Feb 13 23:03:55 ubuntu fail2ban-server[7531]: Server ready
lines 1-14/14 (END)
```

Screenshot : SSH restart success

```
mounika@ubuntu:~$ sudo systemctl restart ssh
mounika@ubuntu:~$
```

7. Incident Timeline

Time	Activity
10:00	Started log monitoring
10:05	Created test user
10:10	Simulated failed login attempts
10:15	Detected suspicious activity in logs
10:20	Classified incident
10:25	Locked user account
10:30	Installed Fail2Ban
10:35	Restarted SSH and verified system

8. Preventive Security Improvements

From this task, I understood the importance of implementing preventive controls such as:

- Strong password policies
- Account lockout mechanisms
- Continuous log monitoring
- Intrusion prevention tools like Fail2Ban
- Multi-factor authentication

9. Conclusion

The simulated security incident involving repeated failed login attempts was successfully detected through authentication log analysis. The incident was classified as a brute force attack attempt, and containment was achieved by locking the affected user account. Preventive measures such as installing Fail2Ban were implemented to reduce future risks. The system was then verified and restored to a secure and stable state after applying the necessary security controls.